# REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

## ARUSHA REGION TERMINAL EXAMS - AUGUST, 2020

### FORM THREE

### BIOLOGY - MARKING SCHEME

## SECTION A : 15 MARKS

1 (i) C
(ii) C
(iii) D
(iv) D
(v) B
(vi) A
(vii) C
(viii) A
(ix) D
(x) D

10 marks @ 01

2 (i) H
(ii) A
(iii) F
(iv) I
(v) C

5 Marks @ 01

## SECTION B : 60 MARKS

3 (a) (i) Bacteriology is the study of bacteria 00½
(ii) Mycology is the study of fungi 00½
(iii) Zoology is the scientific study of animals only. 00½



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3. (b) (i) Students should never handle any chemicals, specimens, apparatus in the laboratory without instructions from the teacher

(ii) A student using any chemical must adhere to the teacher's instructions very carefully to avoid accidents.

(iii) Any accidents involving chemicals or apparatus in the laboratory must be reported to the teacher in charge immediately so that first aid can be admistered.

(iv) In case of accidents, students are advised to be calm. Means panicking may worsen the situation.

(v) Labieled any chemicals or specimens one is using to avoid confusion.

any 3 points @ 01 marks

4. (a) Waste is any unwanted or undesired materials which are no longer needed in the enviroment. 01

(b)(i)(i) It may lead to the environment dirty and unattractive

(ii) It may lead transmission of diseases like cholera

(iii) Decomposed waste gives off unpleasant smell or air pollution.

(iv) When plastic and rubber materials are burnt, the smoke produced causes breathing difficulties and pollutes the air

(v) Death of acquatic organisms

any 3 points @ 01



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4 (b)(ii) (i) Reducing the amount of waste we produce eg. we do not have to buy plastic bags each time we go shopping we can carry a woven basket instead. 01

(ii) Re-using, items instead of throwing them away eg. we can use an empty margarine container to store sugar. 01

(iii) Recycling, materials can be changed to another new products. eg. broken glass can be melted and used to make new glass bottles. 01

5 (a)

| Eukaryotic cell | Prokaryotic cell. |
|---|---|
| (i) The nucleus bounded with membrane | The nucleus not bounded with membrane. |
| (ii) They are found in multicellular organisms eg plants and animals | Mostly found in in uncellular organisms eg bacteria, blue green bacteria |
| (iii) They are bigger in size | They are small in size |
| (iv) They reproduced by either sexually or asexually | They reproduce by binary fission. |
| (v) They have true nuclear cell | They have no true nuclear cell. |

Any 5 points @ 01

5 (b) (i) It performs efficiently and effectively the life processes so that it survives well in its environment.

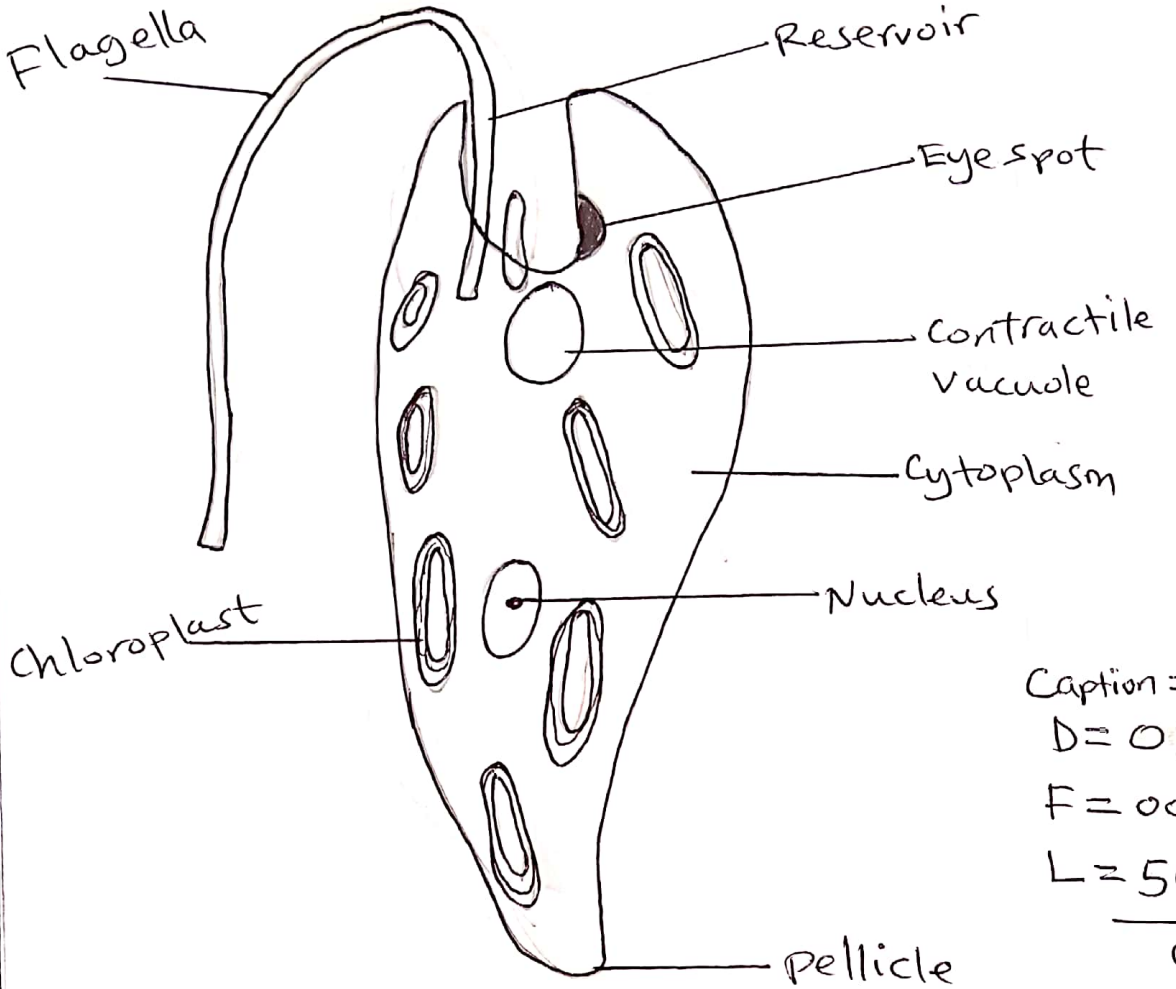
(ii) It lead to cell specialization eg. Cell membrane controls the substances entering or leaving the cell.

any 01 point 01 marks.

Pg3

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6(a) THE STRUCTURE OF EUGLENA.




Caption = 01
D = 01
F = 00½
L = 5@00½
05marks

(b) (i) Euglena 00½

(b) (ii) Economic importance of kingdom Protoctista are.

(i) Some they cause diseases eg. plasmodium cause malaria, Amoeba can cause amoebic dysentery, Paramecium cause balantidiasis.

(ii) They can be used in Laboratories to study cell structures and functions

(iii) Some species used in the Laboratory Culture eg. Some algae.

(iv) Some can be used as a source of food For acquatic organisms eg. Euglena, Paramecium etc.

any 2 @ 01 marks



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7 (a) There will be weak chloroplast and hence photosynthesis processes will stop. 00½

(b)

| Plants | Storage organ. |
|---|---|
| (i) Cassava | Root 00½ |
| (ii) Ginger | Rhizome. 00½ |
| (iii) Onion | bulb 00½ |
| (iv) Irish potato | stem tuber. 00½ |

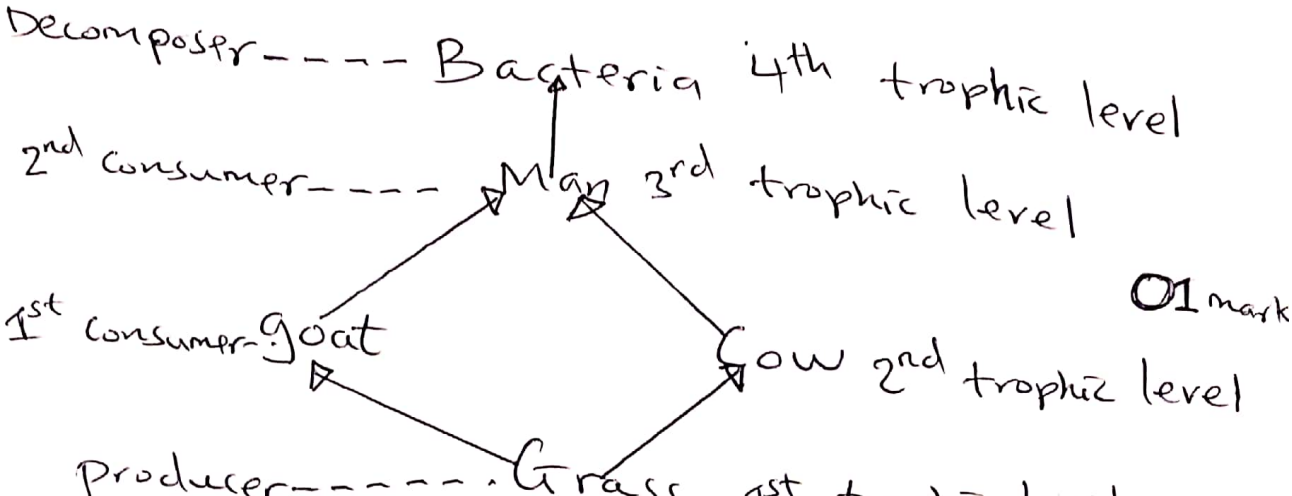
8 (a) (i) Food chain is the sequence of living things in which each organism is the food of the next one in the sequence. 01 marks

(ii) Food web, is the interconnections of many Food chains. 01 marks

(b) (i) Food chains

- Grass → goat → Man → Bacteria 01
- Grass → Cow → Man → Bacteria. 01

(b)(ii) Food web




01 mark



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9 (a) (i) Concentration gradient, a big different between the concentration of the molecules in two regions lead to a faster rate of diffusion.

(ii) Surface area to volume ratio, the higher it is the faster the diffusion rate.

(iii) Distance over which diffusion take place eg: a thin layer of cells increases diffusion rate.

(iv) Temperature

any 03 @ 01 marks

(b) (i) Gaseous exchange in the lungs of animals and the leaves in plants

(ii) Absorption of digested food in the ileum

(iii) Removal of waste materials from the cells and absorption of nutrients and oxygen into the cells.

any 02 @ 01 marks

9 (b) (ii) Amoeba have a contractile vacuole which collects the excess water and removes it from the cells, hence prevents the cell from bursting 01 marks

10 (a) Characteristics of alveolus are.

(i) They are thin to allow easy diffusion gases

(ii) They are numerous to increase the surface area for gaseous exchange

(iii) They are moist to allow easy diffusion of gases.

(iv) They have dense network of blood Capillaries to allow easy transportation of materials.

4 points @ 01 marks



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10 (b)(i) A — Nose
B — Mouth Cavity
C — Trachea
D — Lung. @ 00 1/2

(b)(ii).
(i) Nose has hair like structure and mucus that help to trap dusts and microorganism which can pass through it.
(ii) Presence of mucus enable to warm th moisture air.
(iii) Presence of sensory neurone which — able to detect the type of smell in nose.
⇒ The more appropriate path is nose 00 1/2
any 02 @ 00 1/2

(b)(iii) The effect of regular tobacco smoking Causes lung cancer 00 1/2

11 (a) Tendons, are tough connective tissues — which attach muscles to bones WHILE Ligaments, are fibrous tissue which joints — bone to bone. 02 @ 01

(b) (i) Kidney — Smooth muscles
(ii) Heart — Cardiac muscles
(iii) Alimentary Canal — Smooth muscles
(iv) Legs — Skeletal muscles.
@ 00 1/2



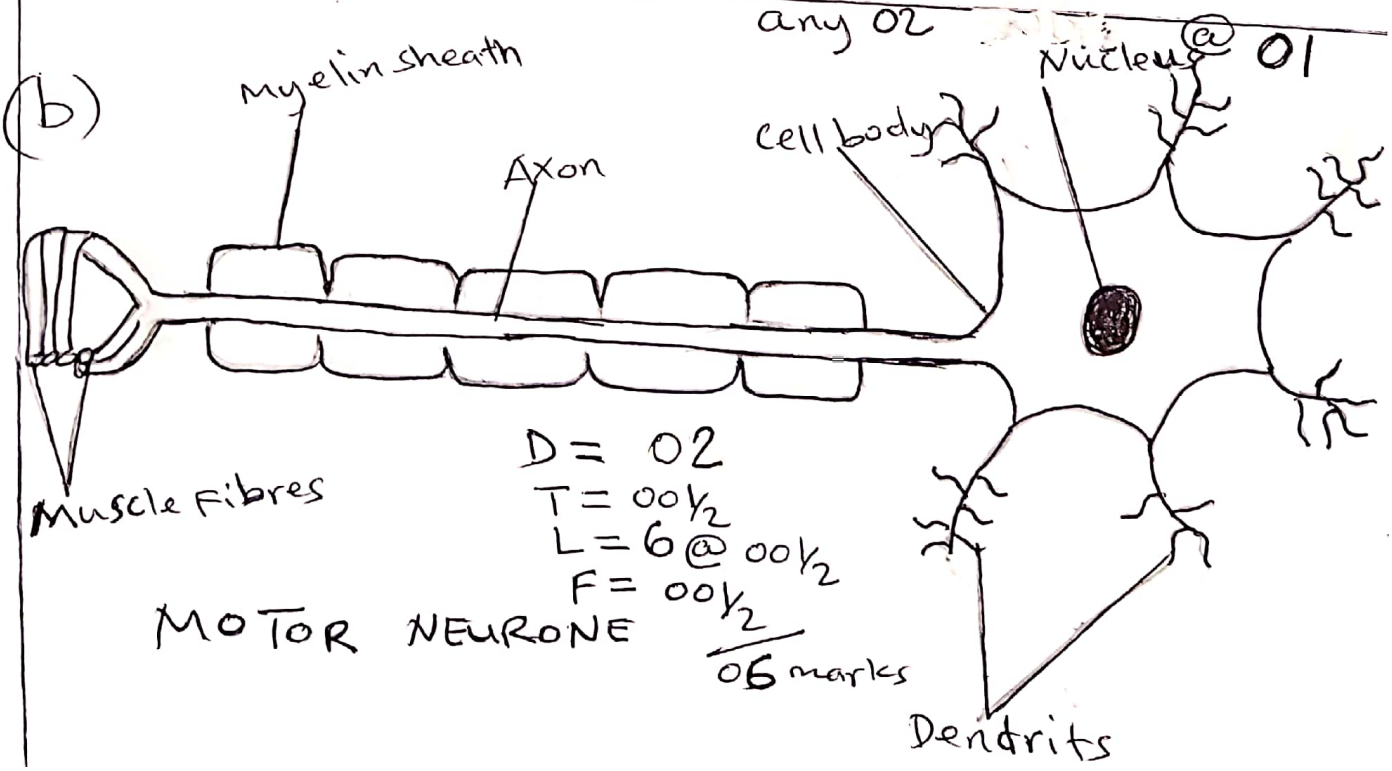
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11 (a)(i) Reflex action is the rapid involuntary response to a certain stimulus. 01 marks

| (a)(ii) Simple reflex action | Condition reflex action. |
|---|---|
| (i) Automatic response which does not depend on learning | Learnt response |
| (ii) Always Constant | - Can be reinforced through rewards or punishment |
| (iii) Response always shown due to specific stimuli | - The same response can be related to varied stimuli |
| (iv) It has a survival value to the stimulus | - It has no survival value to the neutral stimuli |

any 02

(b)




D = 02
T = 00½
L = 6 @ 00½
F = 00½
MOTOR NEURONE
06 marks



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# SECTION C: 25 MARKS

13 (a)(i) Skeleton is a frame work of tissue supporting animal's body. 01

(ii) Types of skeleton.

(1) Hydrostatic skeleton
(2) Exoskeleton
(3) Endoskeleton. @ 00½

Functions of skeleton.

(i) Support, the skeleton provides arigid frame work which supports softer parts of the body eg. provides attachment for muscles and body organs

(ii) Locomotion, the skeleton enables the organisms to move the whole body From one place to another.

(iii) protection, protects delicate internal organs eg. Skull protects the brain, Ribcage protects the lungs, heart etc.

(iv) Formation of blood cells, Foristance Red blood cells and white blood cells are made in the bone marrow.

(v) Shape, the skeleton give animals a definite shape.

any 05 points @ 01

(b) Adaptations of the human skeleton.

(i) presence of bone marrow which synthesized red and white blood cells.

(ii) presence of joints and muscles which brings about movement of an organism. ⟹



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13 (b) (iii) It has firm and strong framework which contains the shape of the body.

(iv) Presence of endoskeleton which is hard and firm.

any 04 points @ 01

Conclusion 01

14 Meaning

COVID-19 is the corona virus disease which is a pandemic disease that can be transmitted from one person to another. 01

Causes

- It is caused by virus known as Corona virus. 01

Transmissions

(i) Through droplet infections during Coughing or sneezing

(ii) Through touching fluids hence contact on face, eyes, nose and mouth. @.02

Symptoms / signs of COVID-19

- Head ache
- Body weakness
- Coughing and sneezing
- fever
- wound on the throat
- Difficult in breathing

any 4 @ 01

Prevention

- Avoid shaking hands and body contact
- Cover nose and mouth during coughing or sneezing.



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14 prevention:

- Avoid touching nose and mouth
- keep, distance of 1 metre to one another
- Wearing Marsk.
- Washing hands frequently by flowing clean water and soap.
- use Sanitizer where necessary.

Conclusion 00½. Any 3 points @ 02 marks

15 Meaning of blood.

- Blood is a fluid which consists of cells. 00½

Components of blood

(i) Red blood cells. mention - - - - 01
Explanation - - - - 01

(ii) White blood Cells mention - - - - 01
Explantation - - - - 01

(iii) plasma. mention - - - - - 01
Explanation - - - - 01

(iv) platelets mention - - - - - - 01
Explanation - - - - - 01

Importance of mammalian blood.

(i) Transportation of gases such as Carbondioxide and oxygen.
(ii) Transportation of digested Foods from the ileum to other parts of the body or body cells
(iii) Transportation of hormones From glands to targeted area.
(iv) Body defence mechanism. eg. White blood cells protect the body against diseases and infections
(v) Distribution of body heat. any 4 @ 01



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